

Sofia (Sonya) Nikiforova

Applied AI/ML Engineer & Data Scientist

NLP · LLMs · 8+ Years Python

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Applied AI/ML engineer and data scientist specializing in NLP, LLMs, and Generative AI, with 8+ years of professional Python experience and a Ph.D. in Computational Linguistics/NLP.

I design, build, evaluate, and improve AI systems that combine language models with structured information, external knowledge, and domain-specific context. My work emphasizes systematic evaluation, error analysis, grounded model outputs, and translating applied AI research into reliable, useful systems.

Experience

Senior AI Engineer

Aug 2026–present

Medendo

Developing AI systems that support mental healthcare professionals: improving quality and reliability through continuous evaluation.

- Designing systematic evaluation workflows for speech-to-text and clinical report generation pipelines, including quality assessment and error analysis in a privacy-sensitive healthcare environment.
- Leading clinical evaluation activities to support MDR certification.

Senior AI Researcher

Oct 2024–June 2026

Causify

Led applied AI research and development spanning LLM-powered data analysis, structured reasoning, and causal inference, with ownership of modeling, evaluation, stakeholder communication, and production-oriented delivery.

- Built an LLM-powered Python tool for generating and validating causal DAGs from datasets and metadata, supporting interactive graph refinement and causal queries backed by formal causal inference.
- Implemented evaluation workflows for LLM-generated outputs, including structural validation, qualitative error analysis, and comparison of outputs across prompts and model configurations.
- Led development of a causal failure prediction framework for wind turbines, combining sensor time series data, domain knowledge, and predictive modeling to identify early indicators of equipment failure.
- Supervised AI and data science interns by defining research objectives, reviewing experimental plans and code, providing regular technical feedback, and assessing final project outcomes.
- Contributed to research publications, technical reports, and patent-related work.

Postdoctoral Researcher

Mar 2023–Oct 2024

Utrecht University

Developed multimodal AI systems combining CT images and radiology reports for clinical decision support in collaboration with AI, medical, and engineering specialists.

- Used locally hosted Llama models via Ollama for NLP-based information extraction from radiology reports, deriving structured clinical context and aligning it with visual findings from medical images.
- Evaluated outputs for factual accuracy, interpretability, and robustness in a safety-sensitive healthcare setting.

Senior Data Scientist

Nov 2016–Jun 2021

Particle.One

Led development of Python-based NLP pipelines for text classification, information extraction, and predictive analytics in the financial domain.

- Owned applied NLP workflows from problem formulation and data acquisition through feature engineering, model training, evaluation, deployment support, monitoring, and iterative improvement.
- Collaborated with business and engineering stakeholders to translate data problems into production-oriented ML solutions.

Teaching Assistant

Sep 2021–Nov 2021

Utrecht University

Developed and graded assignments and supervised practical lab sessions for the MSc course *ML for Human Vision and Language*.

NLP Developer

Jan 2017–Jan 2018

Semantic Hub

Developed biomedical NLP pipelines for entity and relation extraction and structured data generation from user-generated text.

Selected Projects

Contract RAG Agent [GitHub repository](#)

Designed and implemented an agentic RAG system for grounded question answering over contract documents using LangGraph,

Elasticsearch, and FastAPI.

- Built a multi-stage workflow combining evidence retrieval, grounded generation, answer verification, and safeguards against prompt injection and sensitive data disclosure.
- Implemented adaptive recovery paths that revise rejected answers when evidence is sufficient, or rewrite the query and repeat retrieval when it is insufficient.
- Developed an evaluation framework using a gold standard corpus, information retrieval metrics, rubric-based LLM-as-a-judge scoring, and extensive error analysis to improve retrieval and answer quality.

Education

Ph.D., Computational Linguistics/NLP

Utrecht University, Netherlands

2018–Feb 2023

Dissertation: *Image Captioning with External Knowledge*

- Developed transformer-based knowledge-grounded image captioning models in PyTorch.
- Designed and implemented models that identified relevant entities, retrieved contextual information from external knowledge sources, and incorporated that information into generated image descriptions.
- Focused on factual accuracy, grounding, dataset creation, and systematic evaluation across geographic and news-image domains.

M.A., Computational Linguistics

Higher School of Economics, Moscow, Russia

2016–2018

B.A., Fundamental and Applied Linguistics

Lomonosov Moscow State University (MSU), Russia

2012–2016

Selected Publications & Patents

- **DMCD: Semantic-Statistical Framework for Causal Discovery.** *arXiv preprint, 2026.* [arXiv:2602.20333](#)
A two-phase causal discovery framework that integrates LLM-based semantic drafting from variable metadata with statistical validation on observational data.
- **Generating Image Captions with External Encyclopedic Knowledge.** *arXiv preprint, 2022.* [arXiv:2210.04806](#)
Image captioning approach that integrates encyclopedic knowledge into generation, with emphasis on factual accuracy.
- **Geo-Aware Image Caption Generation.** *Proceedings of COLING 2020.* [ACL Anthology](#)
Image captioning with geographic context grounding for generating location-aware photo descriptions.
- **Systems and Methods for Causal Failure Prediction in Time Series** U.S. Patent Application No. 63/934,790

Skills & Technologies

Skills

Natural Language Processing (NLP) · Large Language Models (LLMs) · Generative AI · LLM Evaluation · Retrieval-Augmented Generation (RAG) · Agentic Workflows · Model Evaluation · Error Analysis · Prompt Engineering · Structured Outputs · Information Extraction · Information Retrieval · Dataset Creation · Experiment Design · Explainability · Multimodal AI · Vision-Language Models · Conversational AI · Machine Learning · Causal Inference · Time-Series Modeling

Professional Strengths

Analytical Problem-Solving · Independent Project Ownership · Stakeholder Communication · Cross-Functional Collaboration · Technical Leadership · Agile/Scrum Collaboration

Technologies

Python · OpenAI API · LangGraph · LangChain · Ollama · PyTorch · scikit-learn · pandas · NumPy · Elasticsearch · Pydantic · pytest · Git · GitHub · GitHub Actions · CI Workflows · Docker · FastAPI · SQLite · Jupyter · Linux · LaTeX

Languages

English (fluent) · Dutch (upper-intermediate) · Russian (native)